

Discounting Tax Shields and the Unlevered Cost of Capital

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Abstract: This note derives the relationship between the firm's unlevered cost of capital and its equity cost of capital for a variety of settings. In addition to unifying a number of existing results, a new arbitrage proof is provided which establishes conditions under which the appropriate discount rate of the firm's interest tax shields will equal its unlevered cost of capital. These conditions specify that the firm maintains a *target leverage ratio*, which means that its debt is continuously adjusted according to a target debt-equity (equivalently, debt-to-value) ratio, or to maintain a target interest coverage ratio. This target ratio may vary idiosyncratically over time. Additionally, the firm must have a *horizon independent cost of capital*, which means that its near-term and long-run cash flows have equivalent *current* market risk.

1. Introduction

When a firm is levered, two standard approaches have been put forth to value its cash flows. The weighted average cost of capital (WACC) method discounts the future free cash flows at a cost of capital based on an average of the firm's equity and after-tax debt cost of capital:

$$r_{wacc} = (1-d) r_E + d r_D (1-\tau) \quad (1)$$

where d is the firm's debt-to-value ratio and τ is its tax rate. In this method, the tax advantage of leverage is incorporated by using the firm's debt cost of capital on an after-tax basis.

The second approach is the adjusted present value (APV) method. In this method, the firm's free cash flows are discounted at its unlevered cost of capital, r_U , and the present value of the interest tax shields it will receive are then added separately.

In order for the results of these two methods to coincide, a relationship must exist between the firm's unlevered cost of capital r_U and its equity cost of capital r_E . In this note we determine this relationship and demonstrate its dependence on the type of leverage policy adopted by the firm.

Understanding the relationship between r_E and r_U is not only important for relating the WACC and APV methods. In the APV method, we must use this relationship to estimate r_U based on estimates of r_E and r_D from the firm or its comparables. Even within the WACC method, this relationship is crucial if one wants to compute the effect of a change in leverage on the WACC. If leverage changes, r_E and r_D will also change. While the change to r_D can often be observed, the change to r_E is usually estimated by assuming the underlying risk of the firm, and therefore r_U , remains constant, and then using the relationship between leverage, r_E , and r_U to assess the change to r_E .

Although the relationship between the firm's unlevered and equity cost of capital is crucial for all methods of valuation, there exists a fair amount of confusion in the literature and in practice regarding the correct formula to use. Probably the most used formula is that provided by Modigliani and Miller (1963) in their seminal paper:

$$r_U = \frac{E}{E + D(1 - \tau)} r_E + \frac{D(1 - \tau)}{E + D(1 - \tau)} r_D \text{ or equivalently, } r_E = r_U + \frac{D(1 - \tau)}{E} (r_U - r_D). \quad (2)$$

Versions of these equations were derived for beta by Hamada (1972) and Rubinstein (1973). But while (2) is the most frequently cited result (and appears prominently in many finance texts¹), it was originally derived for a setting in which the firm's debt is permanently held constant, independent of the firm's growth. This assumption is likely to be far from correct for any real firm.

An alternative formula, first developed by Miles and Ezzell (1980, 1985) and Harris and Pringle (1985), is sometimes cited as the appropriate formula when the firm continuously adjusts its debt to maintain a constant leverage ratio. In this formula, the unlevered cost of capital is simply equal to the pretax WACC:²

$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D \text{ or equivalently, } r_E = r_U + \frac{D}{E} (r_U - r_D). \quad (3)$$

Note that (3) is precisely the relationship originally derived by Modigliani and Miller (1958) in a setting without taxes. Because (3) seems to ignore taxes, it is sometimes regarded as incorrect.³

The purpose of this note is two-fold. First, we provide a simple derivation that generalizes (2) and (3) and relates them to an assumption about the market risk, and therefore the appropriate cost of capital, of the firm's tax shields. Here, our results are not new – in addition to the original derivations already cited, recent papers by Arzac and Glosten (2005), Seasholes and Stanton (2005), and Farber, Gillet, and Szafarz (2006) also summarize these results and reach the same conclusions.⁴

- Equation (2) is appropriate if the firm's debt will be permanently held at a fixed level, *and* the firm's tax shields have the same market risk as the firm's debt.

¹ See e.g. Ross, Westerfield, and Jaffe (2005) and Damodaran (2001), which exclusively present the relationship defined by (2) without qualification. Brigham and Houston (2007) also incorrectly apply (2).

² See also Sick (1990), Taggart (1991), Ruback (2002), and Arzac and Glosten (2005).

³ See e.g. Fernandez (2004), whose analysis is critiqued by Fieten, et al (2005).

⁴ For a more comprehensive treatment including the effect of personal taxes, see Berk and DeMarzo (2006).

- Equation (3) applies if the firm's tax shields have the same market risk as the firm's unlevered cash flows.

In addition, we generalize these results to cases in which a portion of the firm's tax shields are predetermined while the remainder share the market risk of the firm's cash flows.

We also present a new, and to our knowledge original, result showing the conditions required for the firm's tax shields to have the same market risk as the firm's unlevered cash flows. In particular, we use a simple arbitrage argument to demonstrate that, given an arbitrary cash flow process for the firm, the following two conditions are sufficient:

- The firm maintains a *target leverage ratio*, meaning that debt is adjusted continuously to maintain either a target debt-equity ratio, or a target interest coverage ratio. Moreover, this target can be allowed to vary idiosyncratically over time.
- The firm's free cash flows have a *horizon independent cost of capital*, meaning that at a given date, the market risk of the firm's short-term and long-term cash flows is equivalent.

This last criterion is a new assumption that is not usually explicitly formulated in the literature. Typically, however, a much stronger assumption is implicitly assumed, which is that the cost of capital associated with any of the firm's cash flows is identical *and constant over time*.

This note proceeds as follows. Section 2 provides basic definitions and the general relationship between the costs of capital of the firm's equity, debt, unlevered cash flows, and interest tax shields. Section 3 rederives (2) and (3) based on different assumptions regarding the risk of the firm's interest tax shields. Finally, Section 4 presents new sufficient conditions for (3) based on the assumption of a target leverage ratio.

2. Definitions and Preliminary Results

Consider a firm that generates free cash flow of c_t at date t . In addition, the firm makes net interest payments of I_t and total payouts to equity holders (dividends plus repurchases less new issues) of ρ_t . If τ is the corporate tax rate and D_t is the value of the firm's net debt at date t , then the budget balancing condition of the firm implies

$$\rho_t = c_t - (1-\tau) I_t + (D_t - D_{t-1})$$

That is, the cash flow paid to equity holders equals the firm's free cash flow less its after-tax interest payments plus increases in its net debt.⁵ To avoid Ponzii schemes, we assume all cash flows cease at some date T and that $D_T = 0$.⁶

⁵ We have assumed for simplicity that all debt is short-term and issued at par, so that $D_t - D_{t-1}$ represents new debt issues net of principal payments, rather than fluctuations in the value of long-term debt. Otherwise, we could denote the value at date t of the debt held at $t-1$ as D_{t-1}^+ , and then net debt issues would be given by $D_t - D_{t-1}^+$. The remaining analysis would be the same.

⁶ A finite horizon is for technical convenience only. One could allow an infinite horizon and impose limits on the growth rate of the firm's debt.

Now let E_t be the ex-dividend value of the firm's equity at date t , so that $V_t^L \equiv E_t + D_t$ is the total enterprise value for the levered firm. Define V_t^U to be the value of an equivalent unlevered firm; that is, the value on date t of a claim to the cash flows c_s for $s > t$. Finally, define T_t to be the value on date t of the firm's future interest tax shields, τI_s for $s > t$.

Given these definitions, we begin with the simple observation that the value of the levered firm must equal the value of the unlevered firm plus the value of the firm's future tax shields:

$$V_t^L \equiv E_t + D_t = V_t^U + T_t, \quad (4)$$

Equation (4) is the basis of the standard APV valuation method. It can be verified inductively by noting that an investor who invests $V_{t-1}^L = E_{t-1} + D_{t-1}$ to hold a portfolio of the firm's debt and equity will have a payoff at date 1 of

$$\begin{aligned} \rho_t + E_t + I_t + D_{t-1} &= c_t + \tau I_t + E_t + D_t \\ &= c_t + \tau I_t + V_t^L \\ &= c_t + \tau I_t + V_t^U + T_t \end{aligned}$$

which is equivalent to the payoff from a portfolio of the unlevered firm and the tax shield security. Therefore, by no arbitrage these portfolios must have the same value at all dates.

Because the portfolios in (4) have equal cash flows, they must also have identical expected returns. Let $r_E(t)$ and $r_D(t)$ be the expected returns of equity and debt at date t , and let $r_U(t)$ and $r_T(t)$ be the expected returns of the unlevered firm and the tax shield security at date t . Then we have⁷

$$E_t r_E + D_t r_D = V_t^U r_U + T_t r_T \quad (5)$$

Equation (5) provides the general relationship between a firm's unlevered cost of capital and its debt and equity cost of capital. It reveals that this relationship will depend upon the expected return, and therefore the risk, of the firm's interest tax shields. The risk of the future tax shields will depend on the firm's leverage policy – in particular, it will depend on how the firm's debt level, and therefore its interest expenses, change in response to changing market conditions. We consider next several standard cases considered in the literature.

3. The Unlevered Cost of Capital

Equation (5) determines the relationship between the firm's equity cost of capital r_E and its unlevered cost of capital r_U . As we can see, this relationship will depend on the risk of the firm's tax shields. Two common assumptions that are used are:

⁷ While the expected returns may vary with t , we suppress the dependence on t unless needed for clarity.

- The firm's debt or interest expenses will increase or decrease with the value or cash flows of the firm, and therefore the interest tax shields have similar risk to the firm's free cash flows.
- The firm's debt or interest expenses are predetermined, and therefore the tax shields have similar risk to the firm's debt.

We consider the consequences of these assumptions below.

Target Leverage Ratio

Suppose a firm continuously adjusts the level of its debt to a *target leverage ratio*, which we define as either a target debt-to-value ratio, or a target ratio of interest to free cash flow. If the firm maintains a target leverage ratio, its future interest tax shields will grow or shrink with the firm. In that case, it is common to assume that the risk of the interest tax shield is similar to the risk of the underlying business, so that $r_T \approx r_U$. (We will look at the conditions needed for a formal justification of this assumption in Section 4.)

If $r_T = r_U$, then Eq. (5) becomes

$$E_t r_E + D_t r_D = (V_t^U + T_t) r_U = V_t^L r_U$$

Rearranging this equation, we see that in this case the firm's unlevered cost of capital is equal to its pretax WACC:⁸

$$\frac{E_t}{E_t + D_t} r_E + \frac{D_t}{E_t + D_t} r_D = r_U \quad (6)$$

Predetermined Debt

When future debt levels are predetermined, the future interest tax shields can be estimated fairly accurately. In this case, it is common to assume that these predetermined or "scheduled" interest tax shields have similar risk to the debt payments themselves, and so the discount rate for the tax shields can be approximated by the firm's debt cost of capital. Suppose the value of the interest tax shields from scheduled debt is T^s , and suppose the remaining value $T - T^s$ of the interest tax shields is from debt that will vary according to a target leverage ratio. Then Eq. (5) becomes

$$E_t r_E + D_t r_D = V_t^U r_U + (T_t - T_t^s) r_U + T_t^s r_D$$

If we let $D_t^s = (D_t - T_t^s)$ be the firm's debt net of the value of its "scheduled" or predetermined tax shields, then we can rewrite this equation as follows:⁹

$$\frac{E_t}{E_t + D_t^s} r_E + \frac{D_t^s}{E_t + D_t^s} r_D = r_U \quad (7)$$

That is, the firm's unlevered cost of capital is equal to its pretax WACC after adjusting the firm's debt to deduct the value of interest tax shields that are already scheduled.

⁸ See Harris and Pringle (1985).

⁹ A similar result is derived in the case $T^s = T$ by Farber, Gillet, and Szafarz (2006).

Intuitively, these predetermined tax shields represent a safe asset which offset the firm's existing leverage.

Special Cases

Equation (7) generalizes Eq. (6) as well as other formulas used in the literature. For example, if the firm's debt level is fixed at D forever, then $T^s = \tau_c D$ and $D^s = (1-\tau_c) D$. Thus, with permanent debt Eq. (7) becomes

$$\frac{E_t}{E_t + (1-\tau_c)D} r_E + \frac{(1-\tau_c)D}{E_t + (1-\tau_c)D} r_D = r_U \quad (8)$$

Equation (8) is the familiar Modigliani-Miller-Hamada formula with permanent debt.¹⁰

Alternatively, suppose the firm's debt is adjusted annually to a target leverage ratio (rather than continuously). Because the interest tax shield for the next year is predetermined, $T_t^s = \tau_c r_D D_t / (1+r_D)$, and Eq. (7) coincides with the results of Miles and Ezzell.¹¹

Leverage and the WACC

We can use the general formula in (7) to determine the relationship between the firm's unlevered cost of capital and its WACC. First, we rewrite (7) to solve for the firm's equity cost of capital:

$$r_E = r_U + \frac{D^s}{E} (r_U - r_D) \quad (9)$$

Using this expression for r_E in the definition of the WACC in (1), we have¹²

$$r_{wacc} = r_U - d\tau[r_D + \phi(r_U - r_D)] \quad (10)$$

where $d = D/(D+E)$, the firm's debt-to-value ratio, and $\phi = T^s/(\tau D)$, which we can interpret as a measure of the permanence of the firm's debt.

Assuming the firm's unlevered cost of capital is unchanged by the change in leverage,¹³ we can also use (10) to compute the impact of a change in leverage on the firm's WACC. Note also that unless d and ϕ are constant over time, the WACC will typically be time-varying.

Equations (7), (9) and (10) thus summarize the relation between the firm's debt, equity, unlevered, and weighted average costs of capital. For practical applications of these formulas to valuation, see Berk and DeMarzo (2006), Chapter 18.

¹⁰ See Modigliani and Miller (1963), and Hamada (1972).

¹¹ See Miles and Ezzell (1985).

¹² Eq. (10) was first developed in Berk and DeMarzo (2006).

¹³ Caution should be exercised when making this assumption. A change in leverage may materially affect the risk of the cash flows due to changes in agency or financial distress costs. See Berk and DeMarzo (2006) for a further discussion, as well as Korteweg (2007) for additional analysis and empirical evidence.

4. Tax Shield Risk with a Target Leverage Ratio

In the previous section we determined the relationship between the firm's equity and unlevered cost of capital given an assumption regarding the risk of the firm's tax shields. While the approximation $r_T \approx r_D$ is natural in the case of pre-determined debt, it is less clear under what conditions the approximation $r_T \approx r_U$ can be justified. In this section we demonstrate that this latter approximation can be justified if the firm maintains a target leverage ratio. First we consider the case in which the firm targets its interest coverage ratio, and then we consider a target debt-equity ratio. In both case we will show that if the market risk of the firm's short-term and long-term cash flows is similar, then r_T can be approximated using the firm's unlevered cost of capital.

4.1. Target Interest Coverage Ratio

Recall that the firm earns free cash flow at rate c_t at date t . Suppose the firm adjusts its debt so that its interest payments equal a target fraction $k(t)$ of this free cash flow. That is,

$$I_t = k(t) c_t.$$

In that case, the firm earns interest tax shields at rate

$$\tau_c I_t = \tau_c k(t) c_t \tag{11}$$

The following result is immediate:

PROPOSITION 1. If the firm maintains a constant interest coverage ratio, so that $k(t) = k$, then $r_T = r_U$ and $V_t^L = (1 + \tau_c k) V_t^U$.

PROOF: The firm's interest tax shields are equivalent to the cash flows from a portfolio of $\tau_c k$ shares of the unlevered firm. ♦

We can extend the result that $r_T = r_U$ to the case in which $k(t)$ is not constant if we make further assumptions regarding the risk of the firm's cash flows. Recall that $r_U(t)$ is the expected return at date t (i.e., from t to $t+1$) of a security that pays cash flows c_s for all $s > t$. Define $r_U(t, t')$ be the expected return at date t of a "truncated" security that pays cash flows c_s for $t < s \leq t'$. Then we say the firm has a **horizon-independent cost of capital** if

$$r_U(t, t') = r_U(t) \text{ for all } t'. \tag{12}$$

Intuitively, this condition states that the current market risk associated with the firm's near term and distant cash flows is equivalent.¹⁴ While this is a strong assumption, it is

¹⁴ That is, we can think of the unlevered firm as a portfolio of near-term and long-term cash flows. The assumption of a horizon-independent cost of capital is that, on a given date, the expected return associated with each of these securities is identical. It is satisfied, for example, if the source of market risk this

an assumption that is often implicitly made in the context of capital budgeting. (In fact, it is often additionally assumed that the unlevered cost of capital does not vary over time, $r_U(t) = r_U$ for all t .)

PROPOSITION 2. If the firm has a horizon-independent cost of capital and maintains an interest coverage ratio given by the function $k(t)$, then $r_T = r_U$.

PROOF: We can construct a security that pays the cash flows in (11) as a portfolio of securities that each promise the cash flow c_s at a different date s , with the function $k(t)$ determining the shares of each security to hold. Under the assumption of a horizon-independent cost of capital, these securities – and thus the portfolio – all have the same expected return as the firm's unlevered equity. ♦

4.2. Target Debt-Equity or Debt-to-Value Ratio

Here we consider a firm that *continuously* adjusts its debt-to-value ratio to a target ratio $d(t)$ on date t , where the schedule $d(\cdot)$ is known in advance. Thus,

$$\frac{D_t}{E_t + D_t} = d(t) \text{ for all } t.$$

(This leverage policy is equivalent to a target debt-equity ratio of $d(t)/(1-d(t))$). First we consider a firm with a single future cash flow, and then a firm with multiple cash flows.

Single Cash Flow

Consider first a firm with a single cash flow c_T paid on date T . The total value at time t of the levered firm is

$$V_t^L = E_t + D_t.$$

The firm adjusts its debt continuously to match a target debt-to-value ratio $d(t)$ so that

$$D_t = d(t)V_t^L \text{ for } s \geq t.$$

Given a corporate tax rate of τ_c , leverage generates an instantaneous interest tax shield of¹⁵

$$\tau_c r_d D_t dt = \tau_c r_D d(t) V_t^L dt \quad (13)$$

period is a multiplicative factor that rescales all future cash flows of the firm, as in standard geometric Brownian motion models.

¹⁵ We assume that the firm can monetize the tax credits associated with net operating losses at date t . Below we consider a firm with multiple cash flows that can use these tax shields to offset the tax liabilities resulting from other income, in which case this assumption is satisfied.

This amount is the combined cash flow generated by the debt and equity together prior to date T . The relationship between the levered and unlevered value of the cash flow can be determined by arbitrage as shown in the following result:

PROPOSITION 3. Given a target debt-to-value ratio $d(\cdot)$, let

$$\alpha_T(t) = e^{\int_t^T -\tau_c r_D d(s) ds}.$$

Then the cash flows from a trading strategy that holds fraction $\alpha_T(t)$ of the equity and debt of the levered firm at each date t are equal to the cash flows from holding the unlevered equity of the firm. Hence, $V_t^u = \alpha_T(t)V_t^L$.

PROOF: From (13), given holdings $\alpha_T(t)$, the trading strategy generates income at rate

$$\alpha_T(t)\tau_c r_D D_t dt = \alpha_T(t)\tau_c r_D d(t)V_t^L dt.$$

Because the cost of increasing our stake in the firm at date t is $V_t^L = E_t + D_t$, the trading strategy requires investment at rate

$$\alpha_T'(t)V_t^L dt = \alpha_T(t)\tau_c r_D d(t)V_t^L dt.$$

Comparing these results, we see that the trading strategy is self-financing, and generates no cash flows prior to date T . On date T , $\alpha_T(T) = 1$, and the trading strategy generates a cash flow of c_T . These cash flows match the cash flows from the unlevered firm. ♦

This result has immediate implications for the relationship between the firm's levered and unlevered cost of capital. Note that the expected return from holding the debt and equity of the levered firm is simply its pretax WACC. Because the unlevered firm is replicated by a portfolio of the debt and equity of the levered firm, the two must have the same expected return – that is, the firm's unlevered cost of capital is equal to the pretax WACC, as in Eq. (6). The interest tax shield is the difference between the levered and unlevered firm,

$$T_t = V_t^L - V_t^u = (1 - \alpha_T(t))V_t^L = \frac{1 - \alpha_T(t)}{\alpha_T(t)} V_t^u. \quad (14)$$

Therefore, the interest tax shield must also have the same expected return r_U .

COROLLARY. Given a target debt-to-value ratio $d(\cdot)$, $r_T = r_U$ and Eq. (6) holds.

We can relate the valuation result of Proposition 3 to the standard WACC method as follows. Suppose that the present value of the unlevered cash flow can be computed by discounting at a known rate $r_U(s)$:

$$V_t^U = e^{-\int_t^T r_U(s) ds} E_t[c_T]$$

Then we have

$$V_t^L = \alpha_T(t)^{-1} V_t^U = e^{-\int_t^T r_U(s) - \tau_c r_D d(s) ds} E_t[c_T] \quad (15)$$

Note that from (6),¹⁶

$$\begin{aligned} r_{wacc} &= (1-d)r_E + dr_D(1-\tau_c) \\ &= (1-d)r_E + dr_D - \tau_c dr_D \\ &= r_U - \tau_c r_D d \end{aligned}$$

Thus, (15) is equivalent to discounting the expected cash flow using the WACC.

Multiple Cash Flows

Now suppose the firm has multiple cash flows, with cash flow c_t paid on date t . Let $V_t^{U,s}$ and $V_t^{L,s}$ be the unlevered and levered value on date t of a security that pays the single cash flow c_s on date s . Standard value additivity results imply that the value of the firm is equal to the total value of each future cash flow:

$$V_t^U = \sum_{s>t} V_t^{U,s} \text{ and } V_t^L = \sum_{s>t} V_t^{L,s} \quad (16)$$

We can use additivity to extend Eq. (14) directly to this setting as follows:

PROPOSITION 4. The tax shield of the levered firm has the same cash flows as a portfolio that on date t holds, for all $s > t$, $\beta_s(t)$ shares of a security that pays c_s on date s , where

$$\beta_s(t) = \frac{1 - \alpha_s(t)}{\alpha_s(t)} = e^{\int_t^s \tau_c r_D d(i) di} - 1$$

If the firm has a horizon-independent cost of capital, then $r_T = r_U$.

PROOF: Note that $\beta_s(t)$ declines to zero as t approaches s , thus the cash flow c_s is not received. Rather, by selling the securities over time, income is generated at rate

¹⁶ We could also compute the WACC by applying (10) with $\phi = 0$.

$$\begin{aligned}
\sum_{s>t} -\beta_s'(t)V_t^{U,s} &= \sum_{s>t} \alpha_s(t)^{-2} \alpha_s'(t)V_t^{U,s} \\
&= \sum_{s>t} \alpha_s(t)^{-2} \alpha_s(t) \tau_c r_d d(t) V_t^{U,s} \\
&= \sum_{s>t} \tau_c r_d d(t) V_t^{L,s} \\
&= \tau_c r_d d(t) V_t^L
\end{aligned}$$

(Note above that we use the fact that $\alpha_s(t)^{-1} V_t^{U,s} = V_t^{L,s}$ from Proposition 3.) Thus, this strategy generates cash flows equivalent to the tax shields of the levered firm.

With a horizon-independent cost of capital, the expected return for all securities in the portfolio, and therefore for the portfolio itself, is r_U . ♦

Note that for a firm with positive net debt ($d(t) > 0$), the number of shares $\beta_s(t)$ in the replicating portfolio increases with the horizon s of the cash flow. Therefore, if the firm's cost of capital is *not* horizon independent, then with a target debt-to-value ratio the expected return of the tax shield will reflect the risk of the firm's more distant cash flows.

5. Conclusion

The primary contribution of this note is a new and simple arbitrage argument to establish conditions under which the firm's interest tax shields should be discounted at its unlevered cost of capital:

- The firm adjusts its debt to maintain a target leverage ratio (i.e., it targets its debt-equity or its interest coverage ratio), and
- The firm has a horizon independent cost of capital (near-term and long-term cash flows have similar *current* market risk).

We also demonstrate that under these conditions, if a portion of the firm's tax shields are predetermined while the remainder vary according to a target leverage ratio, then the general relationship between the firm's unlevered and equity cost of capital is given by

$$r_U = \frac{E_t}{E_t + D_t^s} r_E + \frac{D_t^s}{E_t + D_t^s} r_D \text{ or equivalently, } r_E = r_U + \frac{D^s}{E} (r_U - r_D) \quad (17)$$

where $D^s = D - T^s$, and T^s is the present value of predetermined portion of the firm's interest tax shields, discounted at the firm's debt cost of capital r_D . This formula generalizes the formulas developed by Modigliani-Miller (1963), Hamada (1972), Harris and Pringle (1985), Miles and Ezell (1980, 1985), and others that have appeared in the literature.

6. References

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